

Пробная олимпиада — 4

Time allowed: 3 hours

1. Find all $\alpha \in \mathbb{R}$ for which

$$\lim_{n \rightarrow \infty} n^\alpha \int_0^1 \sin^{2n} \frac{\pi x}{2} dx \in (0, \infty),$$

and compute this limit.

2. Let $a, b \in \mathbb{R}$ be such that $b > a^2$. Find all matrices $A \in M_2(\mathbb{R})$ for which

$$\det(A^2 - 2aA + bI_2) = 0.$$

3. A clock has an hour hand, a minute hand and a second hand, all of length 1. Let T be the triangle formed by the ends of these three hands. A time of day is chosen uniformly at random. Find the expected value of the area of T .

4. Géza has invited his $n \geq 2$ friends to dinner. At his round table there are exactly n seats placed equidistantly, and Géza has put name cards on the seats that completely determine the seating arrangement. Unfortunately, the friends ignored the name cards and just sat down arbitrarily. Depending on n , find the largest positive integer $k(n)$ with the following property: for any seating arrangement Géza can rotate the table so that after the rotation at least $k(n)$ friends are seated correctly. (A rotation by 0 positions is also considered a rotation.)

5. For a positive integer n let $\sigma(n)$ denote the sum of the positive divisors of n . Determine

$$\limsup_{n \rightarrow \infty} \frac{\sigma(n^{2023})}{(\sigma(n))^{2023}}.$$